
WHERE AND WHEN DO BOUNDED ACTIVATIONS HELP INFORMER FORECASTING? FULL-GRID PLACEMENT AND REGIME DIAGNOSTICS UNDER AN OFFICIAL-SCRIPT PROTOCOL

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ABSTRACT

Activation functions inside long-sequence forecasting transformers are usually treated as defaults, yet they can change error without changing attention, depth, embeddings, or parameter scale. We run a full activation-only Informer grid over ETTh1, ETTh2, ETTm1, ETTm2, and eight Solar sites: 36 dataset-horizon cells, nine activation configurations, and three seeds per configuration, for 972 same-protocol seed-runs. All main-table rows use batch size 32, six epochs, fixed horizon-specific Informer windows, and matched GELU baselines. Non-GELU activations are best in 34 of 36 dataset-horizon cells; only Solar3 at horizons 24 and 96 remains GELU-best. Bounded and tanh-family activations rank strongest overall: `tanh_sin`, `tanh`, and `softsign` have the best mean ranks and jointly account for 23 best cells. The gains are largest on ETTm2 and ETTh2, while Solar gains are smaller and site-specific. The full grid also rejects a universal replacement story: output `tanh` wins several high-gain ETT/ETTh cells but is poor on average, ReLU wins only one cell, and factor-oracle diagnostics provide only weak leave-one-seed-out regime evidence. The supported claim is therefore narrow but useful: under a matched Informer protocol, activation choice is a dataset- and horizon-sensitive design variable, not a harmless default.

1 INTRODUCTION

Long-sequence time-series forecasting work usually emphasizes architecture: sparse attention, encoder-decoder shape, temporal embeddings, and horizon-specific training recipes. The nonlinearity inside the feed-forward blocks is often inherited from a default implementation. This is a weak assumption. In a fixed transformer, the activation controls saturation, sign preservation, gradient scale, and the response to large input excursions, all of which can affect forecasting error when the data distribution changes across horizons or regimes.

We study this issue in Informer (Zhou et al., 2021). Informer is a useful substrate because its public scripts specify horizon-dependent configurations for the ETT benchmarks. We keep attention, depth, hidden width, embeddings, optimizer, and training schedule fixed, and change only the activation configuration in hidden blocks or the explicitly marked output-placement variant. The resulting question is deliberately narrow: under the same training environment, does the default GELU-style nonlinearity leave performance on the table?

The answer in our full Phase-4 grid is yes, but not in the form of a universal activation rule. We evaluate 36 dataset-horizon cells across ETTh1, ETTh2, ETTm1, ETTm2, and Solar1–Solar8. Each of the nine activation configurations has three seeds under the same batch-size-32, six-epoch protocol, yielding 972 eligible seed-runs. A non-GELU configuration is best in 34 of the 36 dataset-horizon cells. The strongest average ranks belong to `tanh_sin`, `tanh`, and `softsign`; together they are best in 23 cells. At the same time, the evidence is heterogeneous: Solar3 remains GELU-best at both horizons, output `tanh` wins several ETT/ETTh cells but is poor on average, and ReLU wins only Solar7/24.

This paper makes three contributions.

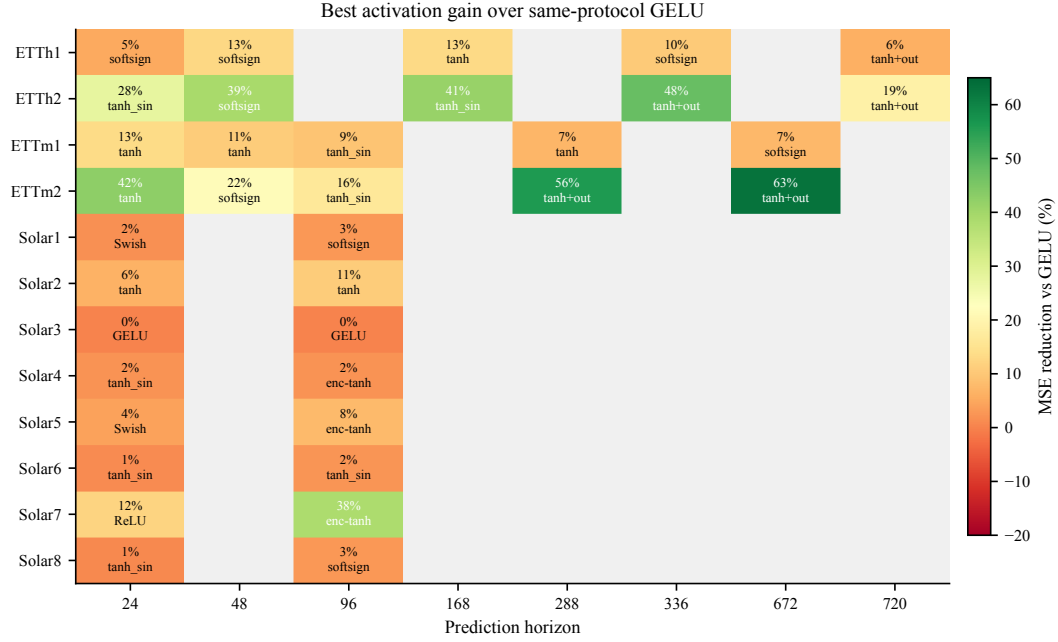


Figure 1: Best activation gain over the same-protocol GELU row in each completed dataset-horizon cell. The full grid covers 36 dataset-horizon cells, nine activation configurations, and three seeds. Most cells prefer a non-GELU activation, but Solar3 remains GELU-best at both horizons.

1. We implement an activation-only Informer interface that switches GELU, ReLU, Swish, bounded activations, a small oscillatory `tanh_sin` variant, encoder/decoder placement, and output activations without changing attention or model width.
2. We report a complete full-grid evaluation with 36 dataset-horizon cells, nine configurations, and three seeds per configuration, reusing exact-matching earlier runs and filling only missing seed-runs.
3. We separate robust same-protocol findings from overclaims: activation choice is dataset- and horizon-sensitive, bounded/tanh-family activations are strong static choices, but no single activation or factor-conditioned router is supported as universally best.

Figure 1 summarizes the main pattern. The rest of the paper describes the controlled activation interface, the protocol, the full-grid results, and the limits of the current evidence.

2 RELATED WORK

Transformers for time-series forecasting. The transformer architecture (Vaswani et al., 2017) introduced attention-based sequence modeling that later became a foundation for long-horizon forecasting architectures. Informer (Zhou et al., 2021) adapts this family to long sequence time-series forecasting through efficient attention and an encoder-decoder design. Our study does not propose a new attention mechanism. Instead, it uses Informer as a fixed experimental substrate to isolate the effect of activation functions.

Activation functions. ReLU (Nair & Hinton, 2010) and GELU (Hendrycks & Gimpel, 2016) are canonical activation baselines for testing nonlinear alternatives. Bounded nonlinearities such as `tanh` and `softsign` are older but still useful controls because they preserve sign while constraining output magnitude. Our experiments show that these bounded controls are essential: without them, a Lee or sinusoidal perturbation may appear useful against GELU while offering little or no advantage over a simpler bounded baseline.

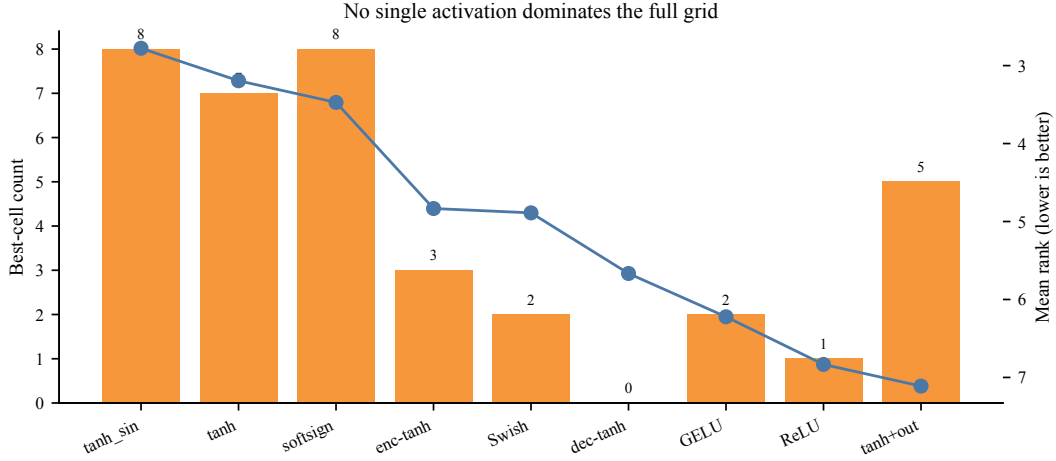


Figure 2: Static activation ranking across the full Phase-4 grid. Bounded and tanh-family activations rank best on average, but the best-cell counts are spread across several configurations, arguing against a universal replacement activation.

Periodic and Fourier-style representations. Periodic activations and Fourier features have been studied for representing high-frequency functions (Sitzmann et al., 2020; Tancik et al., 2020). Our use of oscillatory activations is different. We do not train an implicit neural representation; we perturb the feed-forward activation inside a forecasting transformer and test whether the perturbation improves forecasting error under fixed capacity. This framing makes the comparison against the unperturbed base activation critical.

Regime-conditioned forecasting analysis. Forecasting errors often depend on the state of the input window, such as volatility, trend, anomaly density, or turbulence. We therefore avoid interpreting global MSE alone as a mechanism explanation. Instead, we bind each test sample to input-window factors and ask whether the activation’s relative error improves more in high-factor bins than in low-factor bins. This diagnostic is especially important when every factor bin beats GELU due to a broad bounded-activation effect.

3 METHOD AND EXPERIMENTAL PROTOCOL

3.1 FORECASTING SETUP

Given an input sequence $X_{t-L+1:t} \in \mathbb{R}^{L \times C}$, Informer predicts the next H values $Y_{t+1:t+H}$. We use multivariate forecasting and report mean squared error (MSE) as the primary metric, with mean absolute error (MAE) as a secondary metric. The Phase-4 protocol follows the official-script style used for Informer ETT runs: ProbSparse attention, time-feature embedding, $d_{\text{model}} = 512$, $d_{\text{ff}} = 2048$, 8 heads, batch size 32, learning rate 10^{-4} , patience 3, dropout 0.05, and six training epochs.

The sequence and label lengths are horizon-specific. ETTh cells use horizons 24, 48, 168, 336, 720; ETTm cells use 24, 48, 96, 288, 672; Solar cells use 24, 96. Each horizon uses the corresponding Informer-style sequence length, label length, encoder depth, decoder depth, and ProbSparse factor recorded in the experiment manifest. We call this an official-script protocol rather than an exact reproduction of every number in the original Informer paper.

3.2 ACTIVATION FAMILIES

Let z denote the pre-activation tensor in an Informer feed-forward block. The main hidden activations are

$$f(z) \in \{\text{GELU}(z), \text{ReLU}(z), \text{Swish}(z), \tanh(z), \text{softsign}(z), \tanh(z) + 0.01 \sin(z)\}. \quad (1)$$

Table 1: Phase-4 full-grid coverage. A main-table cell is eligible only when all three seeds are present under batch size 32 and six training epochs.

Group	dataset-horizon cells	config cells	seed-runs	3-seed config cells
ETTh	10	90	270/270	90/90
ETTh	10	90	270/270	90/90
Solar	16	144	432/432	144/144
All	36	324	972/972	324/324

ReLU and Swish are included as non-bounded controls: ReLU tests whether a simpler monotone activation is sufficient, while Swish tests whether smoothness rather than boundedness explains the gains. Output activations are treated as placement variants because they change the final prediction range and can therefore act as target clipping, while hidden activations modify the representation inside the encoder and decoder feed-forward blocks. Lee-style variants and smaller-batch probes are excluded from the Phase-4 main matrix.

3.3 CONTROLLED IMPLEMENTATION

The architecture, attention type, embedding, width, depth, optimizer, and dataset splits are held fixed within each dataset-horizon cell. Activation choices are selected through command-line/config fields and injected into encoder and decoder feed-forward blocks. Placement variants independently set encoder hidden activation, decoder hidden activation, and output activation. This lets us distinguish three hypotheses: a bounded hidden-state effect, a decoder-only effect, and an output-range clipping effect.

3.4 EVIDENCE POLICY

The main tables require batch size 32, six epochs, three seeds, and matched protocol fields. Repeated remote rows are deduplicated by dataset, horizon, seed, protocol, and activation signature before aggregation. Runs with smaller batch sizes, partial seed coverage, non-finite metrics, or mismatched configs are not used to support the main claim. Comparisons to Informer Table 2 are reported only as context: our GELU baseline does not fully reproduce Table 2, so the central claim is same-environment activation improvement over our own GELU runs.

3.5 REGIME DIAGNOSTICS

We additionally compute input-window regime factors such as volatility, realized volatility, volatility shock, max absolute change, tail ratio, jump intensity, trend consistency, reversal rate, and mean-reversion strength. These factors are not used as training labels in the main model. Summary-level factor bins cover all completed dataset-horizon cells. Sample-level same-split and leave-one-seed-out oracles are reported only where prediction arrays are available. Same-split oracles are optimistic diagnostics; leave-one-seed-out results are the stricter evidence for whether a factor is stable enough to guide selection.

4 EXPERIMENTS

4.1 FULL-GRID COVERAGE

The completed Phase-4 matrix contains 972 eligible seed-runs: 36 dataset-horizon cells, nine activation configurations, and three seeds per configuration. All 324 dataset-horizon-configuration cells have complete three-seed coverage under batch size 32 and six epochs. Table 1 summarizes the coverage split across ETTh, ETTh, and Solar.

The aggregation is strict. A row is reusable only when dataset, horizon, sequence length, label length, layer counts, factor, batch size, epochs, activation signature, and seed all match the canon-

ical key. Batch-adjusted diagnostics and mismatched summaries are excluded from the main table. The Solar4 source file contained six missing target values; those values were repaired before re-launching the exact 54 missing Solar4 seed-runs, and the repaired file is recorded by checksum in the experiment audit.

4.2 BEST ACTIVATION BY DATASET AND HORIZON

Table 2 reports the best activation in each completed dataset-horizon cell. A non-GELU activation is best in 34/36 cells. The only GELU-best cases are Solar3 at horizons 24 and 96. ETT and ETTm cells show the largest gains: all 20 ETT/ETTM cells prefer non-GELU activations, with mean best-cell MSE reductions of 22.1% on ETTh and 24.7% on ETTm. Solar gains are smaller and more site-specific: 14 of 16 Solar cells prefer non-GELU activations, with a mean best-cell reduction of 6.7% over GELU among those non-GELU-best cells.

The heatmap in Figure 1 shows the same conclusion visually. Strong positive cells are concentrated in ETTh2 and ETTm2, but the best configuration changes across horizons. `tanh_sin` and `softsign` each win eight cells, `tanh` wins seven, output `tanh` wins five, encoder-side `tanh` wins three, Swish wins two, GELU wins two, and ReLU wins one. This spread is the main empirical reason not to claim a universal activation.

4.3 STATIC RANKING AND PLACEMENT

Table 3 ranks each static configuration within every dataset-horizon cell. The best average ranks are `tanh_sin` (2.78), `tanh` (3.19), and `softsign` (3.47). These three configurations also have high positive-cell rates against GELU, above 80%. Swish is often better than GELU but rarely best, and ReLU is weak overall despite winning Solar7/24.

Placement remains dataset- and horizon-dependent. Output `tanh` is best in five cells, including ETTh1/720, ETTh2/336, ETTh2/720, ETTm2/288, and ETTm2/672. However, its mean rank is 7.11 and its mean relative result is negative because it performs poorly in many other cells. Encoder-side `tanh` is best in three Solar cells, while decoder-side `tanh` wins none. Thus output bounding and encoder placement are real effects, but neither substitutes for a full per-cell activation choice.

4.4 FACTOR-BIN DIAGNOSTICS

Summary-level factor bins cover all 36 completed dataset-horizon cells and produce 783 factor-bin rows. They confirm that activation differences vary with regime summaries, but they do not by themselves prove a deployable selection rule because the bins are computed after observing completed static runs. Sample-level oracles are stricter and are limited to cells with saved prediction arrays. Among 84 leave-one-seed-out oracle rows, only seven are positive, all on Solar1/96. Table 4 lists the positive cases.

The factor evidence is therefore diagnostic rather than causal. Same-split oracles can be read as upper bounds, while the leave-one-seed-out results provide only weak generalization evidence for regime-conditioned selection. The current results support using factors to design targeted follow-up ablations, not deploying a factor-based activation router.

5 CONCLUSION

Activation-only changes materially affect Informer forecasting under the completed Phase-4 official-script protocol. The full grid contains 972 eligible seed-runs and complete three-seed coverage for all 324 dataset-horizon-configuration cells. The strongest supported result is that activation choice is dataset- and horizon-sensitive: non-GELU activations are best in 34 of 36 dataset-horizon cells, and bounded or `tanh`-family configurations have the strongest average ranks.

The evidence also narrows the mechanism. Output `tanh` is not generally beneficial: it wins selected ETT/ETTM cells but ranks poorly on average. Decoder-only `tanh` is weak. Solar gains are smaller and site-specific, and Solar3 remains GELU-best at both tested horizons. Factor-bin and oracle diagnostics show regime dependence, but not a simple rule such as “high volatility always prefers `tanh`.”

Table 2: Best activation per dataset-horizon in the completed Phase-4 grid. Improvements are relative MSE reductions against the same-protocol GELU row with matched seeds; GELU-best cells have no relative-improvement entry.

Dataset	H	best config	MSE	gain vs GELU	seed win rate
ETTh1	24	softsign	0.532 ± 0.026	+5.2%	66.7%
ETTh1	48	softsign	0.646 ± 0.073	+12.6%	100.0%
ETTh1	168	tanh	0.912 ± 0.131	+12.9%	66.7%
ETTh1	336	softsign	1.219 ± 0.042	+10.4%	100.0%
ETTh1	720	tanh+out	1.328 ± 0.030	+6.0%	100.0%
ETTh2	24	tanh_sin	0.516 ± 0.023	+27.8%	100.0%
ETTh2	48	softsign	1.606 ± 0.386	+38.8%	100.0%
ETTh2	168	tanh_sin	2.963 ± 0.509	+41.3%	100.0%
ETTh2	336	tanh+out	2.054 ± 0.219	+47.7%	100.0%
ETTh2	720	tanh+out	3.298 ± 0.186	+18.6%	100.0%
ETTh1	24	tanh	0.314 ± 0.015	+13.4%	100.0%
ETTh1	48	tanh	0.453 ± 0.023	+10.8%	100.0%
ETTh1	96	tanh_sin	0.559 ± 0.050	+9.3%	100.0%
ETTh1	288	tanh	0.802 ± 0.044	+7.2%	66.7%
ETTh1	672	softsign	0.922 ± 0.023	+7.3%	66.7%
ETTh2	24	tanh	0.685 ± 0.191	+42.5%	100.0%
ETTh2	48	softsign	0.309 ± 0.014	+21.6%	100.0%
ETTh2	96	tanh_sin	1.335 ± 0.214	+16.4%	100.0%
ETTh2	288	tanh+out	2.355 ± 0.359	+55.7%	100.0%
ETTh2	672	tanh+out	2.851 ± 0.162	+62.7%	100.0%
Solar1	24	Swish	0.150 ± 0.000	+1.7%	66.7%
Solar1	96	softsign	0.224 ± 0.001	+2.6%	100.0%
Solar2	24	tanh	0.066 ± 0.003	+6.4%	100.0%
Solar2	96	tanh	0.108 ± 0.005	+10.8%	100.0%
Solar3	24	GELU	0.198 ± 0.009	—	—
Solar3	96	GELU	0.349 ± 0.007	—	—
Solar4	24	tanh_sin	0.220 ± 0.003	+1.9%	100.0%
Solar4	96	enc-tanh	0.290 ± 0.005	+1.9%	66.7%
Solar5	24	Swish	0.109 ± 0.002	+4.2%	100.0%
Solar5	96	enc-tanh	0.180 ± 0.003	+7.6%	100.0%
Solar6	24	tanh_sin	0.265 ± 0.002	+1.1%	100.0%
Solar6	96	tanh_sin	0.358 ± 0.009	+2.3%	100.0%
Solar7	24	ReLU	3.281 ± 1.035	+12.0%	66.7%
Solar7	96	enc-tanh	4.970 ± 1.353	+38.2%	100.0%
Solar8	24	tanh_sin	0.216 ± 0.003	+0.8%	100.0%
Solar8	96	softsign	0.306 ± 0.005	+2.6%	100.0%

The next step is not to claim a new universal activation, but to use the completed grid as a map for targeted activation-placement studies. The most defensible design lesson is practical: GELU should not be treated as a harmless default in Informer-style forecasting, but activation choice should be validated per dataset, horizon, and placement rather than promoted as a single global replacement.

Table 3: Static activation ranking across the 36 completed dataset-horizon cells. Rank is computed within each cell over the nine Phase-4 configurations.

Config	mean rank	best cells	top-3 cells	mean gain	positive cells
tanh_sin	2.78	8	27	+11.7%	83.3%
tanh	3.19	7	25	+11.9%	80.6%
softsign	3.47	8	22	+11.2%	80.6%
enc-tanh	4.83	3	11	+5.1%	66.7%
Swish	4.89	2	5	+3.6%	77.8%
dec-tanh	5.67	0	3	+1.9%	66.7%
GELU	6.22	2	4	+0.0%	0.0%
ReLU	6.83	1	4	-2.3%	36.1%
tanh+out	7.11	5	7	-36.3%	25.0%

Table 4: Top positive leave-one-seed-out factor-oracle diagnostics among array-ready Phase-4 cells. These are diagnostic upper bounds for regime sensitivity, not deployed activation-routing results.

Dataset	H	factor	LOSO oracle MSE	best static MSE	gain
Solar1	96	max_abs_change	0.174	0.178	+2.0%
Solar1	96	volatility	0.175	0.178	+1.4%
Solar1	96	volatility_shock	0.175	0.178	+1.3%
Solar1	96	trend_consistency	0.177	0.178	+0.4%
Solar1	96	max_drawdown	0.177	0.178	+0.2%
Solar1	96	max_drawup	0.177	0.178	+0.2%
Solar1	96	range	0.177	0.178	+0.2%

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A ARTIFACT AND REPRODUCIBILITY NOTES

The completed Phase-4 matrix is stored under `phase4_remote_results/analysis_current/`. The aggregate files used by the paper are:

- `phase4_remote_results/analysis_current/phase4_protocol_raw_dedup.csv`
- `phase4_remote_results/analysis_current/phase4_protocol_aggregate.csv`
- `phase4_remote_results/analysis_current/phase4_completion_matrix.csv`
- `phase4_remote_results/analysis_current/phase4_paired_vs_gelu.csv`
- `phase4_remote_results/analysis_current/factors/phase4_factor_bin_summary.csv`
- `phase4_remote_results/analysis_current/factors/phase4_factor_contrast_summary.csv`
- `phase4_remote_results/analysis_current/factors/phase4_factor_oracle_summary.csv`
- `phase4_remote_results/analysis_current/factors/phase4_factor_oracle_loso_summary.csv`

Tables and figures are generated by `figures/gen_phase4_paper_assets.py`. The activation implementation is in `lee_ocil/models/activations.py`; encoder/decoder injection is in `lee_ocil/models/encoder.py` and `lee_ocil/models/decoder.py`. Remote execution scripts are under `lee_ocil/scripts/`.

B DATASET INTEGRITY

The Phase-4 preflight required all 12 local CSV files to exist with the expected schema and target column. Solar4 initially produced non-finite losses. The root cause was six missing values in `Solar/Site_4_130MW.csv`; these were filled using a minimal local policy and the exact Solar4 missing seed-runs were relaunched. The repair audit is stored in `.aris/phase4_solar4_nanfix_20260521.md`, and the repaired Solar4 SHA256 is `47c851e6e04642770cd94e76e5b6782bcf4203fb08cf4695a8d08fa6bbd70672`.

C EXCLUDED DIAGNOSTICS

Batch-adjusted Lee probes and earlier small-batch diagnostics are not part of the Phase-4 main evidence. They are retained in the repository for hypothesis generation, but every claim in the main paper is based on the matched batch-size-32, six-epoch, three-seed matrix.

D CURRENT EVIDENCE LIMITATIONS

The main limitation is that our GELU baseline does not fully reproduce the original Informer Table 2 values. This prevents a claim of reproducing and surpassing the original paper. The present claim is internal and controlled: under the same Phase-4 protocol, activation-only replacements often improve over GELU. A second limitation is that factor-bin results are diagnostic rather than causal. They motivate targeted ablations but do not establish a universal volatility rule or a deployable activation router.